KARE - Project Report Template

Project Title: Detecting Suspicious PowerShell Activity From Windows Logs Using Splunk

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1. Abstract / Project Summary

PowerShell has become one of the most commonly abused tools by attackers due to its deep integration with the Windows operating system and its ability to execute commands in memory. This project focuses on detecting suspicious PowerShell activities by collecting and analyzing Windows logs using Splunk.

The system captures logs such as script execution, process creation, and command-line arguments. These logs are analyzed using Splunk's Search Processing Language (SPL) to identify patterns associated with malicious behavior, including encoded commands, obfuscation techniques, and abnormal execution patterns.

The outcome of this project demonstrates how Security Information and Event Management (SIEM) tools like Splunk can be effectively used to detect real-world cyber threats.

2. Introduction

In modern cybersecurity, attackers often use "living-off-the-land" techniques, where legitimate tools are used maliciously. PowerShell is one of the most widely abused tools in such attacks.

Why PowerShell is dangerous:

1. Already installed in Windows (no need to download malware)
2. Can execute scripts directly in memory (fileless attacks)
3. Supports encoded and obfuscated commands
4. Can bypass traditional antivirus systems

Role of Splunk:

Splunk acts as a **SIEM (Security Information and Event Management)** tool that:

1. Collects logs from systems
2. Indexes and stores them
3. Allows real-time search and analysis
4. Detects suspicious patterns

This project shows how combining Windows logs + Splunk helps detect hidden threats.

3. Objectives

The main objectives of this project are:

- To enable and collect detailed PowerShell logs
- To understand Windows Event Logs related to PowerShell
- To detect suspicious activities using SPL queries
- To analyze attacker techniques such as:
    - Encoded commands
    - Obfuscation
    - Privilege escalation
- To build a basic detection system for real-world cyber threats

4. Tools & Technologies Used

1.Splunk

- Used for log collection, indexing, and analysis
- Supports SPL (Search Processing Language)

2.Windows OS

- Source of logs
- Generates security and PowerShell events

3.PowerShell Logging

Enabled features:

- Script Block Logging (Event ID 4104)
- Module Logging
- Process Creation Logs (Event ID 4688)

4.SPL (Search Processing Language)

- Used to query and analyze logs
- Helps in identifying suspicious patterns

5. Methodology / Steps Performed (Screenshots / Commands)

Step 1: Enable PowerShell Logging

PowerShell logging must be enabled to capture detailed activity.

Types of logging:

- Script Block Logging → Captures full command/script
- Module Logging → Tracks module usage

Command used:

```
Set-ItemProperty
HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging -
Name EnableScriptBlockLogging -Value 1
```

Step 2: Install Splunk Universal Forwarder

- Installed on Windows machine
- Sends logs to Splunk server

Steps:

1. Download Splunk Forwarder
2. Configure receiving indexer
3. Select log sources (Security, PowerShell)

Step 3: Log Collection in Splunk

Important log sources:

- Windows Security Logs
- PowerShell Operational Logs

Key Event IDs:

Event ID Description

4104	PowerShell script execution
4688	Process creation
600	PowerShell engine start

Step 4: Detection Techniques

1. Encoded Commands Detection

Attackers hide commands using Base64 encoding.

```
index=windows EventCode=4104
| search "EncodedCommand"
```

Indicates hidden or obfuscated activity

2. Suspicious PowerShell Execution

```
index=windows EventCode=4688
| search powershell.exe
```

Detects all PowerShell executions
Can be filtered for suspicious parents

3. Long Command Detection (Obfuscation)

```
index=windows EventCode=4104
| eval cmd_length=len(_raw)
| where cmd_length > 500
```

Long scripts often indicate obfuscation

4. PowerShell from Suspicious Parent Process

index=windows EventCode=4688
| search ParentImage="*winword.exe" OR ParentImage="*excel.exe" Indicates possible macro-based attacks

5. Download Cradle Detection

index=windows EventCode=4104
| search "Invoke-WebRequest" OR "DownloadString"

Detects downloading malware from internet

6. Findings / Analysis

From the analysis, the following patterns were observed:

1. Encoded commands are commonly used to hide malicious intent
2. Attackers use PowerShell for:
    - Downloading payloads
    - Executing scripts in memory
3. PowerShell launched from Office apps (Word/Excel) is highly suspicious
4. Long command strings often indicate obfuscation
5. Fileless malware is difficult to detect without log analysis

7. Results

- Successfully configured PowerShell logging
- Logs were collected and indexed in Splunk
- Suspicious activities were detected using SPL queries
- Developed multiple detection rules
- Improved understanding of SIEM-based threat detection

8. Challenges Faced

- Difficulty in enabling advanced logging
- Large volume of logs (noise filtering required)
- Understanding SPL syntax initially
- Differentiating between normal and malicious PowerShell usage

9. Conclusion

This project highlights the importance of monitoring PowerShell activity in Windows systems. Since attackers frequently misuse PowerShell for stealthy attacks, analyzing logs using Splunk provides an effective way to detect threats.

By implementing detection rules and analyzing logs, organizations can significantly improve their security posture and respond quickly to potential attacks.

10. References

- Splunk Official Documentation
- Microsoft PowerShell Logging Guide
- MITRE ATT&CK Framework (T1059.001 – PowerShell)